SOLUTION

RESOLUTION ALGORITHM IMPLEMENTATION

Question 1 – Rain and Wet Ground

1. Propositional Logic

Let:

- **R** = It is raining.
- **W** = Ground is wet.

Knowledge Base:

1. If it rains, the ground becomes wet.
   R → W
2. It is raining.
   R

2. Convert into CNF

The implication:

R → W

is converted as:

¬R ∨ W

Therefore, the clauses are:

- **C1: ¬R ∨ W**
- **C2: R**

3. Negate the Goal

Goal: **W**

Negated goal:

¬W

Add it to the clauses:

- C1: ¬R ∨ W
- C2: R
- C3: ¬W

4. Apply Resolution

Resolve C1 and C3:

(¬R ∨ W), ¬W

Therefore:

¬R

Now resolve:

R, ¬R

Therefore:

□ (Empty Clause)

5. Conclusion

Since the **empty clause (□)** is obtained, the negation of the goal leads to a contradiction.

Therefore, the ground is wet.

Question 2 – Student Assignment Submission

1. Propositional Logic

Let:

- **S** = Rahul submitted the assignment.
- **M** = Rahul receives internal marks.

Knowledge Base:

1. If Rahul submits the assignment, Rahul receives internal marks.
   S → M
2. Rahul submitted the assignment.
   S

2. CNF Conversion

S → M

becomes:

¬S ∨ M

Clauses:

- **C1: ¬S ∨ M**
- **C2: S**

3. Negate the Goal

Goal:

M

Negated goal:

¬M

Clause set:

- C1: ¬S ∨ M
- C2: S
- C3: ¬M

4. Resolution

Resolve C1 and C3:

(¬S ∨ M), ¬M

Result:

¬S

Now resolve C2 and the new clause:

S, ¬S

Result:

□

5. Conclusion

The empty clause is obtained.

Therefore, Rahul receives internal marks.

Question 3 – Library Membership

1. Propositional Logic

Let:

- **L** = Priya is a library member.
- **B** = Priya can borrow books.

Knowledge Base:

1. If Priya is a library member, she can borrow books.
   L → B
2. Priya is a library member.
   L

2. CNF Conversion

L → B

becomes:

¬L ∨ B

Clauses:

- **C1: ¬L ∨ B**
- **C2: L**

3. Negate the Goal

Goal:

B

Negated goal:

¬B

Clauses:

- C1: ¬L ∨ B
- C2: L
- C3: ¬B

4. Resolution

Resolve C1 and C3:

(¬L ∨ B), ¬B

Result:

¬L

Now resolve C2 and the new clause:

L, ¬L

Result:

□

5. Conclusion

The empty clause is derived.

Therefore, Priya can borrow books.

Question 4 – Placement Eligibility

1. Propositional Logic

Let:

- **A** = Arun cleared the aptitude test.
- **P** = Arun is eligible for placement.

Knowledge Base:

1. If Arun clears the aptitude test, he is eligible for placement.
   A → P
2. Arun cleared the aptitude test.
   A

2. CNF Clauses

Convert:

A → P

into:

¬A ∨ P

Therefore:

- **C1: ¬A ∨ P**
- **C2: A**

3. Negate the Goal

Goal:

P

Negated goal:

¬P

Clause set:

- C1: ¬A ∨ P
- C2: A
- C3: ¬P

4. Resolution Step-by-Step

Resolve:

C1: ¬A ∨ P

with

C3: ¬P

Result:

¬A

Now resolve:

C2: A

with:

¬A

Result:

□

5. Resolution Tree

```
¬A ∨ P      ¬P
    \      /
     \    /
      ¬A
       |
```

| A

|

□

6. Conclusion

Since the empty clause is obtained, the negated goal **¬P** is contradictory.

Therefore, Arun is eligible for placement.

Question 5 – Access Control System

1. Propositional Logic

Let:

- **C** = User entered the correct password.
- **A** = User is authenticated.
- **G** = User is granted access.

Knowledge Base:

1. If the user enters the correct password, the user is authenticated.
   C → A
2. If the user is authenticated, the user is granted access.
   A → G
3. The user entered the correct password.
   C

2. Convert into CNF

First implication:

C → A

becomes:

¬C ∨ A

Second implication:

A → G

becomes:

¬A ∨ G

Therefore:

- **C1: ¬C ∨ A**
- **C2: ¬A ∨ G**
- **C3: C**

3. Negate the Goal

Goal:

G

Negated goal:

¬G

Complete clause set:

- C1: ¬C ∨ A
- C2: ¬A ∨ G
- C3: C
- C4: ¬G

4. Apply Resolution

First resolve:

C2: ¬A ∨ G

with

C4: ¬G

Result:

¬A

Next resolve:

C1: ¬C ∨ A

with:

¬A

Result:

¬C

Now resolve:

C3: C

with:

¬C

Result:

□

5. Resolution Tree

```
      ¬A ∨ G        ¬G
          \         /
           \       /
             ¬A
              |
        ¬C ∨ A
           \  /
            ¬C
             |
             | C
             |
             □
```

6. Conclusion

The empty clause □ is obtained. Therefore, the negation of the goal is false, proving the original goal.

Therefore, the user is granted access.